

Measuring the Efficacy of a Low-Cost Tendon-Driven Prosthetic Arm

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Abstract

Affordable prosthetics are becoming an increasingly possible reality using modern technologies to reimagine the principles for making prosthetic limbs. Upper limb prosthetics with multiple degrees of freedom are ideal, as they return functionality to the user after the often traumatic loss of a limb. The design of our anthropomorphic prosthetic implements affordable and accessible parts, such as 3D-printed components, open-source repositories, and consumer-friendly designs to assemble a functioning, posable arm that can be used by amputees. Efficacy will be measured using the GRASP taxonomy, torque output, and tendon durability. Future research includes the implementation of brain-computer interface (BCI) technology to control the device, as well as developing an engineering model for amputee use and testing.

1-Literature Review

1.1 - Importance and Relevance

As of 2017, 57.7 million people across the world experienced limb amputations from traumatic causes including falls, road and transportation injuries, and exposure to mechanical forces (McDonald *et al.*, 2020). More specifically, 564,893 amputation procedures were performed in the United States in 2019. On top of that, limb loss is expected to increase by 145% by the year 2060 due to vascular disease and diabetes (Rivera *et al.*, 2024). With the expected increase in amputations comes the need to replace lost limbs, which leads to the development of prosthetic devices.

Amputees who do not use a prosthetic device experience greater disability and lower physical function in day-to-day life than those who use a prosthesis (Resnik *et al.*, 2020). Active prosthetic devices, or devices that have at least one degree of freedom, are reported to be easier to use by amputees than cosmetic prosthetics, which have no degrees of freedom (Resnik *et al.*, 2020). User and professional reports indicate that requirements for upper limb prosthetics include factors such as weight, battery life, fitment, development of a socket interface, reliability, durability, cleaning capacity, aesthetic appearance, grip strength, and motor functionality (Einfeldt *et al.*, 2023). Another important stipulation is affordability and serviceability to the prosthetic (Einfeldt *et al.*, 2023). The need exists for an active upper limb prosthetic device that is affordable, easy to repair, and capable of a wide range of motor functions.

A large portion of amputees are veterans, many of whom do not receive amputation-related care at the US Department of Veterans Affairs, revealing a large population lacking access to proper care (Resnik *et al.*, 2019). There is substantial evidence pointing to gaps in prosthetic accessibility. Locations such as South and East Asia, and even high-income areas of North America and Western Europe, are found to have limited access to affordable components. The popular prosthetics on the market often lack in functionality or are too expensive to be practical (McDonald *et al.*, 2020). Access to healthcare has been a prevailing issue, most relevantly in rural areas of the U.S., where significant barriers stand in the way of receiving proper health care to people of lower socioeconomic standing and less densely populated areas. (Coombs *et al.*, 2022). All of these factors present a need for low-cost, easily accessible prosthetic devices that amputees around the world, who lack proper access to care, can benefit from. In order to fulfill this need, our project is designed to provide a cost-effective option to those who lack proper access to affordable prosthetic arms.

1.2 - Background Information/Justification for Research

The Anthropomorphic Robotic Arm (ARA) is a proof-of-concept designed to push the field towards creating more affordable and accessible devices for people who have hand and arm disabilities, as part of the broader category of upper limb prosthetic devices discussed previously. According to Hochberg *et al.* (2006), paralyzing disorders profoundly limit independence, mobility and communication. Assistive devices, similar to ARA, are developed to serve for the benefit of the public. The main purpose of this project is to study the effects that low-cost materials have on the mobility and dexterity of prosthetic devices and avoid challenges such as holding, reaching, carrying, and releasing.

Several projects already exist for arm disabilities, and most of them are similar to the anthropomorphic arm. The significant differences between these projects include the controlling systems, materials, durability, and cost. Starting with the controlling systems, there are brain-controlled technological devices called Brain-controlled interfaces (BCI). Brain-controlled interfaces convert the brain signals that provide the movement intentions of the subject into motion in the robotic arm, allowing the subject to become more independent (Arshad *et al.*, 2022). There is a difference between fully accessible BCI and basic brain controlled devices. Basic brain signal-controlled systems are more affordable and more accessible than the alternative. However, they detect the muscle signals coming from the brain, not the exact neuron signal.

Additionally, similar to our own project, Puhlmann *et al.* (2022), developed an anthropomorphic robotic hand with a soft texture actuating finger whose movement functions through pneumatics. The soft nature of the Puhlman hand provides inherent human-like properties where actuation of the joints produces lifelike, accurate movement articulation

(Puhlmann *et al.*, 2022). A key feature we see not only in this hand but also in others is an opposing thumb (Puhlmann *et al.*, 2022). This allows for articulation of motion more representative of a real human hand which is able to move the thumb across the palm while also bending it like a normal finger while also being able to touch each finger on the hand. This group utilizes the Kapandji test for thumb opposition, which serves as a litmus test for a given thumb's opposition that we could implement in our own internal testing methodology (Puhlmann *et al.*, 2022). Furthermore, they tested their hand capabilities of performing all 33 types of grasps, known as the GRASP Taxonomy (Puhlmann *et al.*, 2022), providing yet another test for which our hand can be benchmarked.

To separate ourselves not only from prior designs of this specific project, but the entire field of robotic arms, we intend to make a few key changes in this iteration. Notably, being the utilization of a planetary robotic actuators, inducing high strength and definition of elbow motion, a task previously inaccessible to us. In regard to the wrist, we will utilize a direct-drive servo to actuate one degree of freedom, providing an additional kinematic pathway while keeping the design efficient. The planetary robot actuators will take input from a stepper motor mounted at a 90-degree angle, then perform a reduction, increasing the torque produced greatly. This will then output to an elbow joint, which can produce motion by the input passed by the stepper motor through the planetary gearbox. The torque provided by the gearbox is an essential component in this instance, allowing for the long lever arm and moderate weight of the arm to be manipulated by the singular input stepper motor.

2-Experimental Design

2.1 - Introduction

Limb amputation has unfortunately had an increasing prevalence rate with the contributions of traumatic injury, vascular disease, and diabetes (McDonald *et al.*, 2020). Many who have this struggle are unable to acquire the correct resources they need to fully function in day-to-day life and consequently, are faced with disability (Resnik *et al.*, 2020; Einfeldt *et al.*, 2023). There is a large number of amputees who simply do not have access to healthcare because of the expense and the cost of prosthetic devices (Einfeldt *et al.*, 2023). As a result, we aim to create an anthropomorphic robotic arm with the goals of achieving regular mobility and affordability for amputees. Our project prioritizes mobility, affordability, control, and durability as the leading factors for the model.

2.2 - Design Brief

2.2.1 Objective

Our objective for designing this prosthetic arm was to develop a proof-of-concept made affordably with hobbyist materials while maintaining a low weight, dexterity, and high elbow torque output that could be classified and published such that others would be inspired to continue our research, pushing the fields of biomedical engineering and robotics towards the development of cheaper and more accessible prosthetics.

2.2.2 Criteria

- Maximize hand dexterity
- Minimize prototype weight
- Minimize cost as much as possible
- Generate large torque proportional to the product weight
- Maximize precision/repeatability

2.2.3 Constraints

- Achieve 18/33 grasp types when measured against the GRASP Taxonomy
- Produce a Torque output of 10 Nm at the elbow
- Do not exceed \$300 in material costs

2.3 System Design

2.3.1 Materials and Manufacturing

To manufacture the arm, we will utilize FDM 3D printers exclusively for their accessibility, low cost, as well as their capacity for rapid prototyping. Parts will be printed in PLA and PETG. We will use PLA for the parts that require less strength, such as the fingers and forearm shell, as they will not bear load in this case. For the parts that will bear a heavy load and will be run through more cycles, such as the gearboxes or mount, we will use PETG. To produce higher strength and better tolerances, we will tune the infill density and infill pattern, as well as the contour compensation and hole compensation settings in the slicer.

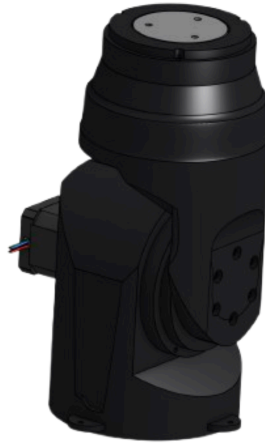
2.3.2 Actuation

2.3.2.1 Servos

Servo motors will be used in our project to control the fingers and joints of the ARA. Standard servos are used for the wrist joint, and micro servos for each finger. The fingers are threaded using fishing line and attached to the servo horn. This allows the servo to control the flexion and extension of each finger individually, as well as splaying. The servos in the wrist will allow for rotating and bending of the arm, adding two degrees of freedom to the ARA.

2.3.2.2 Stepper Motors

Stepper motors are our solution of choice for driving the planetary gearboxes that actuate the elbow's pitch and roll axes, as they are affordable yet provide precise control.



2-DOF Elbow Joint

2.3.3 Transmission

2.3.3.1 Tendons

To translate the rotational output of the servos perpendicularly about their axes, we utilize fishing line to emulate tendons. These ‘tendons’ are attached to the servo's output shaft as a pulley, and run through the fingers off-center towards the palm, curling the fingers close when the pulley tightens the fishing line about itself. When the fishing line is unwound, the elastics in the back of the fingers impart restorative force, returning the fingers to their open state.

2.3.3.2 Planetary Gearboxes

To support the force of the arm and be able to produce a high-definition rotational output at the elbow, we needed to find a way to increase the torque of the stepper motor. To do this, we chose a planetary gearbox, as it provides a means to increase torque in a smaller package while also being largely unaffected by our manufacturing methods relative to other reducers. For our elbow pitch, we are constrained to produce at least 10 Nm of torque. The gearbox is actuated by a NEMA-17 Stepper motor that produces a torque of 84 oz·in, being reduced as it passes

through a 64:1 planetary gearbox. At 100% efficiency, that yields roughly 38 N·m of torque. Due to the difficulty 3D Printers suffer from when printing gears, as well as the friction caused by plastic on plastic rotation and the lack of bearing, a harshly pessimistic assumption of efficiency would be 50%. This would still yield 14 N·m of torque, well above the targeted 10 N·m of torque.

64:1 Planetary Gearbox



2.3.4 Power

Since we need 1 standard servo for the wrist, 5 standard servos for the fingers, 5 micro servos for the finger splaying/thumb opposition, and 2 stepper motors for the elbow, a significant amount of power is needed for this project to operate. Thus, due to the high power draw of the servos, a power supply to match these demands is necessary.

2.3.4.1 Power Supply

Choosing a suitable power supply for this project was an important part of the design process as the team has faced issues powering all the components in the past. We decided to use a lithium polymer (LiPo) battery, similar to a lithium ion (Li-ion) battery, which is able to store a lot of energy, last for a long time, and hold a charge longer than other batteries (Shahbazi *et al*, 2026). The LiPo battery provides more versatility than the Li-ion batteries; since it is made of a liquid electrolyte. This feature allows the battery to be packaged in a more flexible, lightweight casing, making it ideal for projects with limited space (Shahbazi *et al*, 2026).

The exact battery we are utilizing is the Tattu G-Tech 3700mAh 4S 45C LiPo Battery Pack. The battery capacity required was found by determining the highest voltage requirement of the project, the total current draw, and desired runtime, using the following equation:

$$\text{Capacity (mAh)} = \text{Total current (mA)} \times \text{Desired runtime (hours)}.$$

Figure 2. Capacity Calculation

The maximum continuous current of the battery is determined by multiplying the battery capacity in amp hours (Ah) and the C-rating of the battery.



Figure 3. Tattu G-Tech 3700mAh 4S 45C LiPo Battery Pack

2.3.4.2 Buck Converter

Since the battery outputs 14.8 Volts, which is suitable for the stepper motor drivers but not the servos and other electronics, we need to reduce the voltage down to 5V. To achieve this, we are going to utilize a buck converter, which takes an input of a higher voltage and drops it down to the desired voltage, in this case, 5V.

2.3.5 Electronics & Control

2.3.5.1 ESP32 Microcontroller

An ESP32 microcontroller is used to control the arm. This component will allow us to program the arm as well as allow communication between the arm and any modes of control.

2.3.5.2 PWM Signaling (PCA9685)

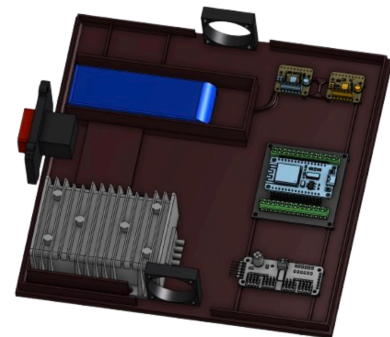
Instead of using one io pin on the esp32 to drive each servo, we will be using a PCA9685 board, which uses I2C communication, taking two pins from the esp32 to open up the PCA9685 board to send up to 16 unique PWM signals.

2.3.5.3 Stepper Control

The stepper motors that drive the planetary robotic actuators in the elbow require dedicated driver boards to function. Our ESP32 outputs two signals to each driver board: the step and dir signals. The STEP signal controls movement, where each 'STEP' makes the stepper motor move one increment. The DIR pin sets the direction in which the motor rotates.

2.3.5.4 Electronics Housing / Mounting Plate

To mount the arm, we need to develop a base plate that is strong enough to support its weight, while also concentrating enough weight in the base such that it shifts the center of mass down, preventing the arm from falling over in motion. To strengthen the mounting plate, we will add ribs to the underside of its top face to support a larger amount of force and manufacture it out of PETG. To lower the center of mass, we will house the electronics in the bottom of the mounting plate, additionally providing a location to ventilate and cool the electronics while keeping them stationary, reducing the risk of damage.



2.3.6 Software and Control

To control the arm, an ESP32 in the electronics housing hosts a portal accessible over its own network. In this portal, there are sliders controlling each individual axis in the arm, including the 11 servo motors controlled on the PCA9685 board as well as the stepper motors driven by the TMC2209 and A4988 stepper driver boards, respectively. The firmware also has a calibration button that sets the arm to the same position at the beginning and end of each program, as well as a manual zeroing function that creates virtual encoders for the stepper motors.

2.4 - Methods

2.4.1 - GRASP Taxonomy Assessment

We plan to test the dexterity of our arm using the GRASP taxonomy developed by Feix *et al.* (2016). Since no proper consensus on range of common human grasp types exists, their study synthesized the current existing taxonomies of human grasps into a taxonomy of 33 static and stable grasps arranged according to opposition type, virtual finger assessments, type in terms of power, precision, or intermediate grasp, and position of the thumb. Feix *et al.* (2016) define a grasp as every steady hand grip that can hold an object securely regardless of permutation of said hand. They further specify that regardless of the relative force direction between the object and the hand, the stability of the grasp must be maintained.

The taxonomy can be reduced to a set of 17 or more general grasps when not considering the object shape or size (Feix *et al.*, 2016). We plan to test our device against all 33 grasp types, but we are aiming for 18 out of the 33 grasps as a metric for success.

2.4.2 Measuring Dexterity

To measure the dexterity of our hand, we will assess it against the GRASP Taxonomy developed by Feix *et al.* (2016). For each grip, the hand will have to perform a ten-second hold

without slipping on a uniquely shaped object to simulate the gesture, with success being measured as such. In the end, we will yield a score out of 33, representing hand dexterity.

2.4.3 Defining Torque Output

To capture the output torque of the elbow, we will fasten the arm to a test fixture, holding the bicep joint at a 90-degree angle parallel to the ground. We will then attach a cable going from the load cell to a fixed point at a fixed distance from the stepper motor, perpendicular to the forearm. We will then begin the test by increasing the force imparted by the stepper motor, with the load cell measuring the force exerted until the motor can no longer support the load being placed on it. The maximal force output of the elbow joint will be recorded at the moment just before the motor slips or stalls. This data will then be converted to Torque using the formula

$$\tau = F \cdot r \cdot \sin \Theta$$

After which, we will reference the maximal force output to determine if we achieved our goal of 10 N·m torque output, as well as graphing the torque output of the actuator over time.

2.4.4 Tendon Durability Testing

To test the durability of the tendons running through the fingers, we will write a script that runs a ‘cycle’ or one open-close cycle of a finger curling, and increments each time it does so until the program is stopped. If a tendon can achieve 10,000 cycles without failure, it will have passed the durability test, otherwise, it will be documented.

2.4.5 Arm Classification

The three benchmarks previously discussed: dexterity, torque output, and durability, all produce quantifiable data that can be compiled into a classification for any given individual robotic arm, including GRASP types achieved, torque output in N·m, and pass/fail on a 10,000-cycle tendon durability test. This will not only allow us to publish results for others to

observe our data, but also provide a testing methodology for others attempting to build on our research to compare their arm against ours through standardized means.

2.5 - Per Unit Cost

The final all-inclusive per-unit cost of our prototype is \$290.54. This price complies with our constraint from section 2.2.3 to develop the arm at a cost of under \$300, showing that at face value, the proof-of-concept prosthetic arm can be built in that price range utilizing consumer-grade hardware.



2.6 Grant Request

To collect reproducible data and provide a benchmark for others to compare and build on our research, we require the following equipment, each contributing to the data collection in the areas of dexterity, torque, or durability.

Grant Request			
Equipment	Purpose	Minimum	Maximum
S-Type Tension Load Cell, 0-500 N, with Digital Indicator	Measure Torque Output at the Elbow	\$90	\$210
Aluminum Extrusions	Build a Test Fixture	\$120	\$260
Mass Set	Calibration of the Load Cell	\$40	\$110
Documentation Equipment	Document tests / trials	\$150	\$320
Second Arm	Utilize a brand new arm for testing purposes only	\$290	\$290
Total Request		\$690	\$1190

This funding would allow us to produce research impacting the fields of biomedical engineering and robotics by bringing attention to the merit of low-cost prosthetic devices.

3-Possible Outcomes

Hypothesis Introduction

As mentioned in the previous section, the arm will be classified by its dexterity and torque output. We will base our hypothesis upon these qualities.

Primary Hypothesis:

If 18 out of 33 GRASP taxonomy types or more (Feix *et al.*, 2016) exhibit passing criteria and 10 N·m sustained elbow torque is achieved, then the effect of using cheaper materials is conducive to a successful prosthetic device that can then be further developed for different amputee needs.

Grip force is high on the importance of prosthetic characteristics for amputees surveyed by Henao *et al.* (2025), as well as the ability to perform everyday activities, such as housework, outdoor activities, and personal care and grooming. Another important expectation expressed by users and their relatives is that the prosthetics should allow them to move all their fingers independently (Henao *et al.*, 2025), which our current model is able to do. If the dexterity of our device can enable users to be able to perform a significant portion of their daily activities at a much cheaper cost as well as better meet their expectations, it would stand to reason that this device has great potential to be a successful prosthetic for users.

Mentioned previously, an example of a successful hand, able to perform all 33 grasp types of the GRASP taxonomy (Feix *et al.*, 2016), is the RBO Hand 3 developed by Puhlmann *et al.* (2022). With a similar material cost to our device (less than \$250), the RBO Hand 3 has 16 degrees of freedom, whereas our “ARMin” has 13. Since both hands are relatively similar in

regards to degrees of actuation, it is reasonable to hypothesize that our hand should be able to perform at least half of the grasps in the taxonomy.

Alternative Hypothesis:

If less than 18 out of the 33 grasp types exhibit passing criteria, but 10 N·m sustained elbow torque is achieved, the device will be moderately successful, with a rework of only the mechanics of the hand required. This outcome would mean that our arm has the strength to perform most daily tasks, but lacks the dexterity to properly grip or hold things.

Users are more likely to abandon their prosthetics if their prosthetic lacks dexterity and they are unable to complete their daily activities with the device (Henao *et al.*, 2025). If our arm is not able to perform a majority of the grasp types, it is more likely to be abandoned by the user, and thus would not be a successful prosthetic device.

Since the RBO Hand 3 is pneumatically actuated (Puhlmann *et al.*, 2022), whereas ours is tendon-driven, our hand could fail to match the dexterity of a soft robotic hand like this one. It is possible that our tendon system could face difficulties and wear through use over time, which is why we will also be testing cycles until failure as well.

Null hypothesis:

If neither the required number of grasps nor the sustained elbow torque is achieved, then this would mean that our device fails to perform its intended functionality and will require a full rework in order to achieve these success metrics for a viable prosthetic device. In this case, not achieving either benchmark would most likely lead to user abandonment, since the arm would be unable to perform enough, if any, everyday activities that users require (Henao *et al.*, 2025).

4-Future Directions

Brain Control Interface (BCI)

As more advanced technology becomes more and more accessible to the public, we can integrate better equipment into the arm with a lower expenditure. One device that we think would greatly progress the ability of the arm is a Brain-Computer Interface (BCI) . This would allow the user to move their prosthetic through the input of their personal brain waves. BCI utilizes an electroencephalogram (EEG) to read the waves a brain produces. Every minuscule movement made produces a wave on the EEG. A computer is able to interpret and translate these waves into executable motions made from the user to the prosthetic (Arshad *et al.*, 2022). If this was able to be made for practical daily use, BCI could transform how consumers interact with the health field.

Mounting harness/socket interface for amputee testing

Fitment and comfort are often cited as contributing reasons as to why amputees abandon their prosthetics (Henao *et al.*, 2025). Perfecting a comfortable and practical socket and mounting harness will be a key aspect into commercial success of open-source robotic arms. Our current project is not a wearable object. We intend to modify this design to be a functioning prosthetic. This would open the door for real-world testing opportunities and data with which we could adjust for the things that the theory doesn't cover.

Sensory Feedback Systems

A large issue that many users face is the lack of responsive data they receive from their prosthetic devices. In order to properly maneuver through daily life, a person must use their sense of touch to measure things like how much force to apply or how far they should move their

fingers. This restriction could be greatly reduced by adding devices that would send the somatic data from the hand back to the user.

Flex sensors are already a popular product in the healthcare field. Used in wearable architecture, as well as embedded in clothing (Saggio, 2012). Flex sensors are made of a thin conductive material whose voltage shifts when bent, causing a shift in resistance at various angles. With the proper data, we would be able to track the movements of a person's fingers and send that information back to the user in real time so that they would be able to orient themselves accordingly (Saggio 2012).

5-References

- Arshad, J., Qaisar, A., Rehman, A. U., Shakir, M., Nazir, M. K., Rehman, A. U., Eldin, E. T., Ghamry, N. A., & Hamam, H. (2022). Intelligent control of robotic arm using brain computer interface and artificial intelligence. *Applied Sciences*, 12(21), Article 10813. <https://doi.org/10.3390/app122110813>
- ARA 2024 [unpublished, Baladi *et al.*]
- ARA 2025 [unpublished, Baez *et al.*]
- Coombs, N. C., Campbell, D. G., & Caringi, J. (2022). A qualitative study of rural healthcare providers' views of social, cultural, and programmatic barriers to healthcare access. *BMC Health Services Research*, 22(1), 438. <https://doi.org/10.1186/s12913-022-07829-2>
- Einfeldt, A., Rebmann, F., Yao, D., Stukenborg-Colsmann, C., Hurschler, C., Windhagen, H., & Jakubowitz, E. (2023). What do users and their aiding professionals want from future devices in upper limb prosthetics? A focus group study. *PLoS One*, 18(12), e0295516. <https://doi.org/10.1371/journal.pone.0295516>
- Feix, T., Romero, J., Schmiedmayer, H., Dollar, A. M., Kragic, D. (2016). The GRASP taxonomy of human grasp types. *IEEE Transactions on Human-Machine Systems*, 46(1), 66-77. <https://doi.org/10.1109/THMS.2015.2470657>
- García, N., Suárez, R., & Rosell, J. (2017). First-order synergies for motion planning of anthropomorphic dual-arm robots. *IFAC-PapersOnLine*, 50(1), 2247-2254. <https://doi.org/10.1016/j.ifacol.2017.08.187>
- Henao, S. C., Cuartas-Escobar, S., Salazar-Salgado, S., Posada-Borrero, A. M. (2025). Upper-limb prosthetic requirements from the healthcare providers, end-users and

relatives' perspectives. *Journal of Hand Therapy*, 39, 147-155.

<https://doi.org/10.1016/j.jht.2025.04.006>

Hochberg, L. R., Serruya, M. D., Fiehs, G. M., Mukand, J. A., Saleh, M., Caplan, A. H., Branner, A., Chen, D., Penn, R. D., & Donoghue, J. P. (2006). Neuronal ensemble control of prosthetic devices by a human with tetraplegia. *Nature*, 442(7099), 164–171.

<https://doi.org/10.1038/nature04970>

McDonald, C. L., Westcott-McCoy, S., Weaver, M. R., Haagsma, J., & Kartin, D. (2022). Global prevalence of traumatic non-fatal limb amputation. *Prosthetics and Orthotics International*, 00(0), 1-12. <https://doi.org/10.1177/0309364620972258>

Nag, A., Menzies, B., Mukhopadhyay, S. C. (2018). Performance analysis of flexible printed sensors for robotic arm applications. *Sensors and Actuators, A* 276, 226-236.

<https://doi.org/10.1016/j.sna.2018.04.031>

Puhlmann, S., Harris, J., & Brock, O. (2022). RBO Hand 3: A platform for soft dexterous manipulation. *IEEE Transactions on Robotics*, 38(6), 3434–3449.

<https://doi.org/10.1109/TRO.2022.3156806>

Resnik, L., Borgia, M., & Clark, M. (2020). Function and quality of life of unilateral major upper limb amputees: Impact of prosthesis use and type. *Archives of Physical Medicine and Rehabilitation*, 101(8), 1396-1406. <https://doi.org/10.1016/j.apmr.2020.04.003>

Resnik, L., Ekerholm, S., Borgia, M., & Clark, M. A. (2019). A national study of Veterans with major upper limb amputation: Survey methods, participants, and summary findings. *PLoS One*, 14(3), e0213578. <https://doi.org/10.1371/journal.pone.0213578>

- Rivera, J. A., Churovich, K., Anderson, A. B., & Potter, B. K. (2024). Estimating recent US limb loss prevalence and updating future projections. *Archives of Rehabilitation Research and Clinical Translation*, 6(4), 100376. <https://doi.org/10.1016/j.arrct.2024.100376>
- Saggio, G. (2012). Mechanical model of flex sensors used to sense finger movements. *Sensors and Actuators A*, 185, 53-58. <http://doi.org/10.1016/j.sna.2012.07.023>
- Shahbazi, M., Seidi, E., & Ferreira, A. (2026). Comparative analysis and PSO-based optimization of battery technologies for autonomous mobile robots. *Batteries*, 12(3), 108. <https://doi.org/10.3390/batteries12030108>
- Varanis, M., Silva, A., Mereles, A., & Pederiva, R. (2018). MEMS accelerometers for mechanical vibrations analysis: A comprehensive review with applications. *Journal of the Brazilian Society of Mechanical Sciences and Engineering*, 40(11), 527. <https://doi.org/10.1007/s40430-018-1445-5>